GenAI

UNIT 1 Hands-On 1 Observations
Generative Artificial Intelligence & NLP Basics

Part 1: Overview of Hugging Face and the Transformers Library

What is Hugging Face?

Hugging Face is a popular open-source platform that allows researchers and developers to share and access pre-trained artificial intelligence models. It serves as a central hub for modern NLP models, making it possible to experiment with advanced architectures without the need for heavy computational resources or long training times. Models such as GPT-2, BERT, and RoBERTa can be directly downloaded and used for experimentation.

What is the transformers library?

The transformers library provides the necessary tools to interact with models hosted on Hugging Face. It offers simple and unified APIs to load, configure, and run transformer-based models. This library supports multiple deep-learning frameworks, allowing users to work in environments such as PyTorch or TensorFlow without changing their approach.

What is the pipeline() function?

The pipeline() function is a high-level utility that simplifies NLP workflows. It automatically manages:

- Converting raw text into tokens
- Passing tokens through the model
- Translating model outputs into readable results

As a result, complex NLP tasks can be executed with very few lines of code.

Observations

- Hugging Face makes advanced AI models easily accessible.
- The transformers library hides low-level implementation details.
- The pipeline() function is well-suited for rapid experimentation and learning.

Task 1: Seed Value Experiment

Objective

The aim of this task was to study how randomness influences text generation in language models by modifying the seed value and observing the changes in output.

Experiment Configuration

- **Model Used**: GPT-2
- **Input Prompt**:
  "Generative AI is a revolutionary technology that..."
- **Seed Values Tested**: 42 and 68

Output (Seed = 42)

```
Device set to use cuda:0
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly tru
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the
Generative AI is a revolutionary technology that enables a wide range of intelligent systems to work independently from one another.
```

Generative AI is a revolutionary technology that enables a wide range of intelligent systems to work independently from one another.

Generated Output (Seed = 68)

```
Device set to use cuda:0
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples
Setting `pad_token_id` to `eos_token_id`:50256 for open-end generation.
Both `max_new_tokens` (=256) and `max_length`(=50) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation
Generative AI is a revolutionary technology that is enabling us to imagine what it's like to be human and what it might take to become a social for

The researchers believe their technology will help overcome the challenges of technology that are becoming increasingly difficult to overcome.

"It's a very exciting opportunity," said Eric Fierstein, an associate professor of computer science at the University of Pennsylvania who was not i

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Generative AI is a revolutionary technology that is enabling us to imagine what it's like to be human and what it might take to become a social force," he said. "It's not just that we can be AI, or it could be something else entirely."

The researchers believe their technology will help overcome the challenges of technology that are becoming increasingly difficult to overcome.

"It's a very exciting opportunity," said Eric Fierstein, an associate professor of computer science at the University of Pennsylvania who was not involved in the research. "It's a very exciting time for the field, and a very exciting time for the future of the world."

The research was presented at the Society for Artificial Intelligence (SAAI) Conference on Artificial Intelligence and the Future of Social Robotics, held Nov. 20-22 in Toronto, Canada.

Observations

- Keeping the same seed produces identical output across runs.
- Changing the seed value leads to noticeable variation in generated text.
- Seed 42 results in a more direct and informative response.
- Seed 68 produces a more abstract and reflective continuation.
- This confirms that text generation is probabilistic in nature.

Key Takeaway

Seed values play an important role in controlling randomness during text generation. Even with the same model and prompt, altering the seed can significantly change the output.

Task 2: Comparative Study of BERT, RoBERTa, and BART

Objective

The objective of this task was to compare multiple transformer models across different NLP tasks and understand how architectural design affects their performance.

Models Considered

BERT

- Encoder-only architecture
- Trained using Masked Language Modeling

RoBERTa

- Improved and optimized version of BERT
- Trained with larger datasets

BART

- Encoder-Decoder architecture
- Trained to reconstruct corrupted input sequences

Experimental Results

Task	Model	Classification	What happened?	Why did this happen? (Simple Explanation)
Generation	BERT	Failure	Generated only dots / meaningless output	BERT only understands text; it is not built to generate new text
Generation	RoBERTa	Failure	Repeated or echoed the input	Like BERT, RoBERTa cannot predict the next word
Generation	BART	Failure	Generated incoherent or low-quality text	BART can generate text, but this version is not trained well for free text generation
Fill-Mask	BERT	Success	Predicted *create, generate, produce*	BERT is trained to guess missing words
Fill-Mask	RoBERTa	Success	Predicted *generate, create, discover*	RoBERTa is trained even better at guessing missing words
Fill-Mask	BART	Success	Predicted *create, help, provide*	BART is trained to fix corrupted or missing text
Question Answering	BERT	Partial	Extracted *"such as hallucinations"*	The model understands the sentence but is not trained to pick exact answers
Question Answering	RoBERTa	Partial	Extracted *", bias, and"*	Same problem as BERT — no training for answer selection
Question Answering	BART	Partial	Extracted *"and deepfakes."*	BART is not meant for selecting exact answer spans

```
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EXPERIMENT 1: TEXT GENERATION
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1. BERT:
config.json: 100%|| 570/570 [00:00<00:00, 61.8kB/s]
model.safetensors: 100%|| 440M/440M [00:02<00:00, 278MB/s]
If you want to use `BertLMHeadModel` as a standalone, add `is_decoder=True.`
tokenizer_config.json: 100%|| 48.0/48.0 [00:00<00:00, 3.10kB/s]
vocab.txt: 100%|| 232k/232k [00:00<00:00, 3.34MB/s]
tokenizer.json: 100%|| 466k/466k [00:00<00:00, 7.82MB/s]
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first
Both `max_new_tokens` (=256) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.c
[{'generated_text': 'The future of Artificial Intelligence is.....................................................................................................

2. RoBERTa:
config.json: 100%|| 481/481 [00:00<00:00, 27.8kB/s]
model.safetensors: 100%|| 499M/499M [00:02<00:00, 291MB/s]
If you want to use `RobertaLMHeadModel` as a standalone, add `is_decoder=True.`
tokenizer_config.json: 100%|| 25.0/25.0 [00:00<00:00, 955B/s]
vocab.json: 100%|| 899k/899k [00:00<00:00, 17.1MB/s]
merges.txt: 100%|| 456k/456k [00:00<00:00, 12.0MB/s]
tokenizer.json: 100%|| 1.36M/1.36M [00:00<00:00, 13.0MB/s]
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first
Both `max_new_tokens` (=256) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.c
[{'generated_text': 'The future of Artificial Intelligence is'}]
```

```
3. BART:
config.json: 1.72k/? [00:00<00:00, 99.5kB/s]
model.safetensors: 100% 558M/558M [00:04<00:00, 257MB/s]
Some weights of BartForCausalLM were not initialized from the model checkpoint at facebook/bart-base and are newly initialized: ['lm_head.weight', 'model.decoder.embed_tokens.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
vocab.json: 899k/? [00:00<00:00, 15.3MB/s]
merges.txt: 456k/? [00:00<00:00, 15.9MB/s]
tokenizer.json: 1.36M/? [00:00<00:00, 25.2MB/s]
Device set to use cpu
Truncation was not explicitly activated but `max_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting to 'longest_first
Both `max_new_tokens` (=256) and `max_length`(=20) seem to have been set. `max_new_tokens` will take precedence. Please refer to the documentation for more information. (https://huggingface.c
[{'generated_text': 'The future of Artificial Intelligence is inefficientarthed Frey681681 inadvertUserritoCAP681ibly terrorismiblyiblyibly cooldown cooldown Wol681681 latitude Wolibly Nidibl
```

```
Some weights of the model checkpoint at bert-base-uncased were not used when initializing BertForMaskedLM: ['bert.pooler.dense.bias', 'bert.pooler.dense.weight', 'cls.seq_relationship.bias',
- This IS expected if you are initializing BertForMaskedLM from the checkpoint of a model trained on another task or with another architecture (e.g. initializing a BertForSequenceClassificat
- This IS NOT expected if you are initializing BertForMaskedLM from the checkpoint of a model that you expect to be exactly identical (initializing a BertForSequenceClassification model from
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EXPERIMENT 2: FILL-MASK
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1. BERT:
Device set to use cpu
Top 3: ['create', 'generate', 'produce']

2. RoBERTa:
Device set to use cpu
Top 3: [' generate', ' create', ' discover']

3. BART:
Device set to use cpu
Top 3: [' create', ' help', ' provide']
```

```
Some weights of BertForQuestionAnswering were not initialized from the model checkpoint at bert-base-uncased and are newly initialized: ['qa_outputs.bias', 'qa_outputs.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
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EXPERIMENT 3: QUESTION ANSWERING
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1. BERT:
Device set to use cpu
Some weights of RobertaForQuestionAnswering were not initialized from the model checkpoint at roberta-base and are newly initialized: ['qa_outputs.bias', 'qa_outputs.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Answer: such as hallucinations

2. RoBERTa:
Device set to use cpu
Answer: , bias, and

3. BART:
Some weights of BartForQuestionAnswering were not initialized from the model checkpoint at facebook/bart-base and are newly initialized: ['qa_outputs.bias', 'qa_outputs.weight']
You should probably TRAIN this model on a down-stream task to be able to use it for predictions and inference.
Device set to use cpu
Answer: and deepfakes.
```

Observations

- Encoder-only models are ineffective for text generation.
- BERT and RoBERTa perform best in masked word prediction.
- BART requires task-specific training to generate meaningful text.
- Question answering tasks need fine-tuned models for accuracy.

Key Takeaway

A model's architecture and training objective directly determine which tasks it can perform well. Using a model outside its intended purpose results in poor performance.

Task 3: AI-Based Spell Check and Grammar Correction System

Objective

The objective of this task was to implement a practical NLP application that corrects grammatical and spelling errors using contextual understanding rather than rule-based methods.

Project Overview

The spellcheck system identifies incorrect words by masking them and uses BERT to predict suitable replacements based on surrounding context. This approach allows for grammar-aware corrections instead of simple dictionary matching.

Tools and Technologies

- **Model Used**: bert-base-uncased
- **Task Type**: Fill-Mask
- **Library**: Hugging Face Transformers

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AI GRAMMAR & SPELL FIXER

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Original: I am [MASK] to school tomorrow.
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Suggestions:
  1. I am going to school tomorrow. (confidence: 91.4%)
  2. I am off to school tomorrow. (confidence: 3.2%)
  3. I am heading to school tomorrow. (confidence: 1.6%)


Original: She have [MASK] the book yesterday.
----------------------------------------------------------
Suggestions:
  1. She have read the book yesterday. (confidence: 36.1%)
  2. She have seen the book yesterday. (confidence: 3.6%)
  3. She have done the book yesterday. (confidence: 3.4%)


Original: The cat is [MASK] on the mat.
----------------------------------------------------------
Suggestions:
  1. The cat is sitting on the mat. (confidence: 14.9%)
  2. The cat is lying on the mat. (confidence: 9.9%)
  3. The cat is sleeping on the mat. (confidence: 8.1%)


Original: He don't [MASK] his homework.
----------------------------------------------------------
Suggestions:
  1. He don't do his homework. (confidence: 85.3%)
  2. He don't have his homework. (confidence: 3.3%)
  3. He don't finish his homework. (confidence: 1.9%)
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Original: They was [MASK] happy.
------------------------------------------------------------
Suggestions:
  1. They was so happy. (confidence: 37.2%)
  2. They was not happy. (confidence: 17.8%)
  3. They was all happy. (confidence: 9.6%)


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TRY IT YOURSELF!
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Replace the word you want to check with [MASK]
Example: 'I am [MASK] to the store'

Testing: The weather is [MASK] today.

Top 3 suggestions:
1. The weather is good today.
2. The weather is warm today.
3. The weather is mild today.
```

Observations

- BERT produces accurate grammatical corrections when the error is clearly masked.
- High confidence scores indicate strong contextual understanding.
- The system performs well for common grammatical structures.
- Manual identification of incorrect words is required.

Limitations

- Automatic error detection is not supported.
- Only one word can be corrected at a time.
- Performance depends on sentence clarity and context.

Final Key Takeaway

This set of experiments demonstrates a complete learning progression—from understanding randomness in text generation, to evaluating model capabilities, and finally applying this knowledge to build a useful AI-based language tool. Selecting the correct model for a specific task is essential for achieving reliable results in Generative AI applications.